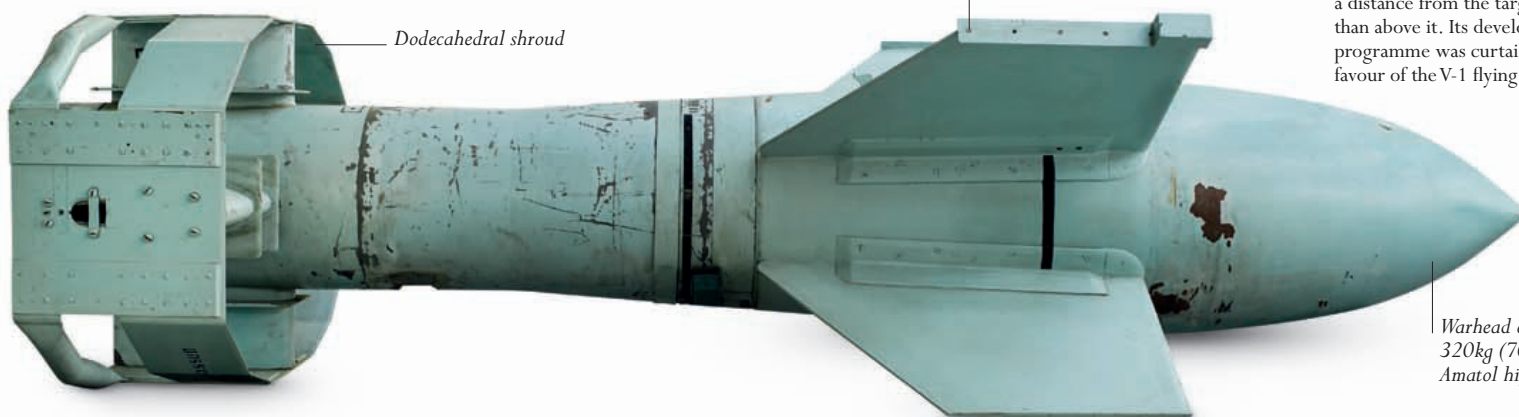


BOMBS AND MISSILES OF WORLD WAR II

Initially, aerial bombs were simple iron cylinders filled with explosives and fused to detonate on impact; sometimes they had rudimentary fins, but were still only suitable for hitting areas rather than specific targets. Aerodynamic casings and more effective fins made flight paths more predictable, but even the best bombs were far from accurate. As soon as appropriate technologies became available, bombs were fitted with guidance systems, some of them built-in, others requiring a controlling hand in the launch aircraft. By 1944, following much experimentation with rocketry, the first generation of “surface-to-surface” missiles appeared (i.e. launched from the ground to strike ground targets), and promised to change the very nature of the war in the air.



▲ RUHRSTAHL/KRAMER X-1 “FRITZ X”

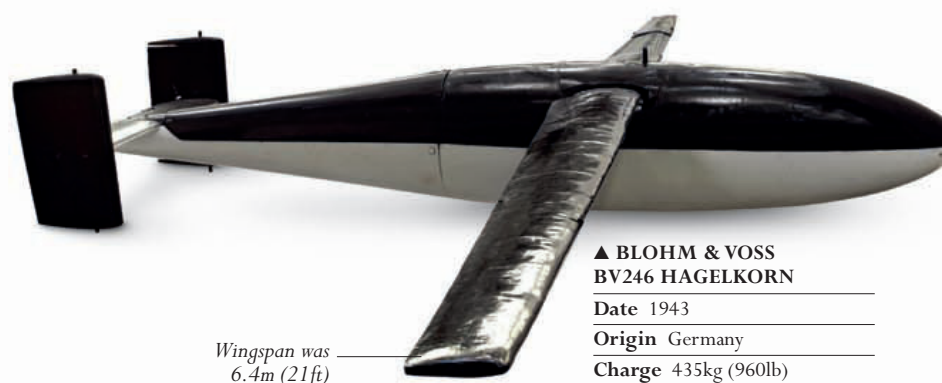
Date 1943

Origin Germany

Charge 320kg (705lb)

Length 3.3m (11ft)

The most successful guided bomb of the war, the “Fritz X” was steered to its target by an operator aboard the launching aircraft. It was a marked success, with the *Luftwaffe*’s specialist unit, *Kampfgeschwader 100*, sinking the British ship *HMS Warspite* at Salerno.



▲ BLOHM & VOSS BV246 HAGELKORN

Date 1943

Origin Germany

Charge 435kg (960lb)

Length 3.5m (11½ft)

The *Hagelkorn* (“Hailstone”) glide-bomb was an unpowered air-to-surface missile, its gliding ability allowing it to be dropped at a distance from the target rather than above it. Its development programme was curtailed in favour of the V-1 flying bomb.

Wingspan was 6.4m (21ft)

Stabilizing fins

Warhead contained 320kg (705lb) of Amatol high explosive

▼ FIESLER FI103 (V1)

Date 1944

Origin Germany

Charge 830kg (1,830lb)

Length 8.32m (27¼ft)

The first long-range surface-to-surface missile, the FI103 was launched from a ramp by a steam catapult, and was powered by a simple pulse-jet motor. It was deployed from 13 June 1944 against targets in the UK and later against Antwerp.



Ranging device

Warhead contained 850kg (1,874lb) of Amatol high explosive

► HENSCHEL HS 117H SCHMETTERLING

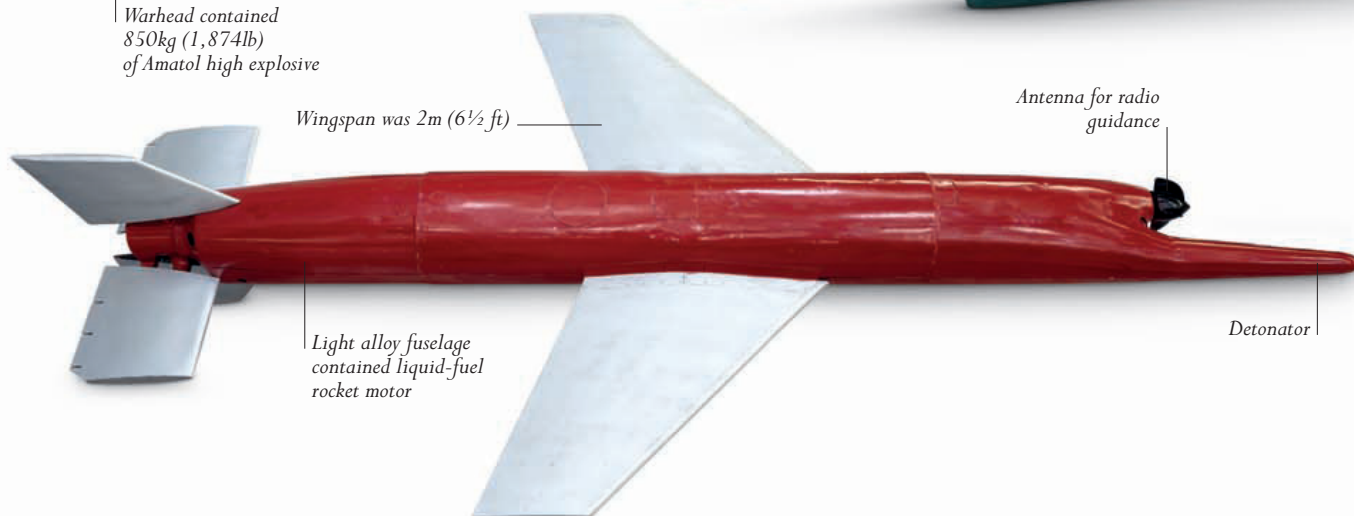
Date 1944

Origin Germany

Charge 25kg (55lb)

Length 4.2m (13¾in)

The Henschel *Schmetterling* (Butterfly) was a surface-to-air and air-to-air radio-guided anti-aircraft missile. Its latter form, the HS 117H, dispensed with the rocket boosters fitted to the surface-to-air missile variant. Its maximum range was 10km (6¼miles), and it travelled at 940kph (585mph).



Wingspan was 2m (6½ft)

Antenna for radio guidance

Light alloy fuselage contained liquid-fuel rocket motor

Detonator